

## WINDOM, MN, USA

WMO: 727506

Lat: 43.913N

Lon: 95.109W

Elev: 1410

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04952

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -11.2      | -7.9 | -18.4                         | 2.1 | -10.8 | -14.7 | 2.6 | -7.0 | 34.2                  | 13.1 | 32.0 | 15.4 | 8.4                      | 320  | 0.616 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.9                   | 90.2            | 74.0 | 87.8 | 72.8 | 84.1 | 71.3 | 78.2                | 85.7 | 76.2 | 83.5 | 74.3 | 81.2 | 13.2                 | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.4                            | 140.1 | 82.4 | 73.3 | 130.4 | 80.6 | 72.2 | 125.7 | 79.5 | 42.7          | 86.4 | 40.7 | 83.1 | 38.7 | 81.5 | 93.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 29.0          | 26.0 | 23.4 | DB                         | -19.0 | 95.4               | 8.2 | 5.0                                                | -25.0 | 99.0       | -29.8 | 102.0      | -34.4 | 104.8      | -40.4 | 108.5 | (4) |
| (5)               |      |      | WB                         | -19.6 | 82.4               | 7.9 | 3.6                                                | -25.3 | 84.9       | -29.9 | 87.0       | -34.3 | 89.0       | -40.1 | 91.6  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 45.9   | 16.4  | 19.0  | 32.4  | 46.2  | 57.9  | 68.2  | 72.7  | 69.4  | 62.5  | 48.9  | 35.2  | 20.5  |
|                                                                                  | DBStd       | 22.40  | 13.05 | 13.01 | 13.30 | 10.75 | 9.24  | 6.76  | 5.72  | 5.76  | 8.75  | 9.93  | 11.97 | 12.29 |
|                                                                                  | HDD50       | 4224   | 1042  | 870   | 562   | 194   | 30    | 0     | 0     | 0     | 9     | 141   | 461   | 915   |
|                                                                                  | HDD65       | 7641   | 1507  | 1289  | 1012  | 569   | 258   | 43    | 8     | 23    | 149   | 509   | 894   | 1380  |
|                                                                                  | CDD50       | 2728   | 0     | 0     | 16    | 79    | 275   | 545   | 703   | 601   | 385   | 107   | 17    | 0     |
|                                                                                  | CDD65       | 670    | 0     | 0     | 1     | 4     | 37    | 138   | 246   | 159   | 75    | 10    | 0     | 0     |
|                                                                                  | CDH74       | 6224   | 0     | 0     | 14    | 73    | 433   | 1267  | 2319  | 1292  | 705   | 118   | 3     | 0     |
|                                                                                  | CDH80       | 1937   | 0     | 0     | 6     | 17    | 140   | 407   | 783   | 348   | 206   | 30    | 0     | 0     |
| Wind                                                                             | WSAvg       | 10.2   | 11.5  | 11.7  | 11.4  | 12.1  | 11.3  | 9.3   | 7.5   | 7.0   | 8.8   | 10.1  | 11.1  | 10.9  |
| Precipitation                                                                    | PrecAvg     | 30.8   | 0.7   | 0.7   | 1.8   | 3.2   | 3.8   | 4.7   | 4.0   | 3.9   | 3.3   | 2.2   | 1.4   | 0.9   |
|                                                                                  | PrecMax     | 42.2   | 2.2   | 1.5   | 4.1   | 7.5   | 7.4   | 10.2  | 6.3   | 8.8   | 12.3  | 6.3   | 3.9   | 2.6   |
|                                                                                  | PrecMin     | 22.7   | 0.0   | 0.2   | 0.0   | 0.6   | 1.2   | 2.4   | 1.3   | 1.0   | 0.6   | 0.4   | 0.1   | 0.2   |
|                                                                                  | PrecStd     | 5.6    | 0.5   | 0.4   | 1.0   | 1.7   | 1.6   | 1.8   | 1.4   | 2.1   | 2.9   | 1.5   | 1.1   | 0.7   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 46.1  | 57.2  | 72.5  | 81.8  | 90.1  | 92.8  | 93.3  | 90.7  | 90.1  | 84.0  | 70.4  | 52.3  |
|                                                                                  |             | MCWB   | 39.8  | 48.9  | 57.2  | 61.2  | 66.5  | 72.8  | 78.1  | 75.9  | 70.8  | 65.4  | 55.7  | 46.7  |
|                                                                                  | 2%          | DB     | 39.2  | 46.5  | 63.5  | 74.5  | 83.7  | 88.3  | 90.3  | 87.7  | 85.5  | 76.7  | 63.3  | 45.1  |
|                                                                                  |             | MCWB   | 35.3  | 42.2  | 53.9  | 56.8  | 64.6  | 71.2  | 76.5  | 75.6  | 69.4  | 61.0  | 52.9  | 39.7  |
|                                                                                  | 5%          | DB     | 36.5  | 39.5  | 56.7  | 69.9  | 79.2  | 84.2  | 87.8  | 83.8  | 81.7  | 71.5  | 57.2  | 39.0  |
|                                                                                  |             | MCWB   | 34.2  | 36.5  | 50.1  | 55.0  | 62.6  | 69.2  | 75.2  | 73.1  | 67.4  | 58.8  | 49.0  | 35.8  |
|                                                                                  | 10%         | DB     | 34.1  | 36.4  | 51.5  | 63.8  | 73.2  | 81.5  | 84.0  | 81.2  | 78.6  | 64.3  | 52.4  | 36.3  |
|                                                                                  |             | MCWB   | 32.4  | 34.3  | 47.5  | 52.1  | 59.6  | 67.7  | 73.6  | 71.2  | 66.0  | 54.8  | 46.2  | 34.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 40.3  | 51.3  | 61.3  | 65.0  | 71.5  | 78.1  | 82.2  | 79.9  | 75.5  | 71.4  | 59.3  | 47.3  |
|                                                                                  |             | MCDB   | 46.3  | 56.1  | 67.1  | 75.9  | 83.1  | 86.2  | 89.0  | 85.9  | 83.6  | 77.0  | 66.1  | 49.5  |
|                                                                                  | 2%          | WB     | 36.7  | 42.3  | 55.2  | 60.0  | 67.7  | 75.2  | 79.1  | 77.3  | 73.3  | 63.9  | 54.8  | 39.9  |
|                                                                                  |             | MCDB   | 38.8  | 46.9  | 61.1  | 70.1  | 77.9  | 83.7  | 86.8  | 84.5  | 81.2  | 70.5  | 61.8  | 44.0  |
|                                                                                  | 5%          | WB     | 34.4  | 37.7  | 51.0  | 56.8  | 65.0  | 72.7  | 77.2  | 75.0  | 70.9  | 59.9  | 50.3  | 36.7  |
|                                                                                  |             | MCDB   | 36.3  | 40.1  | 56.0  | 66.8  | 74.8  | 80.3  | 84.7  | 81.3  | 78.6  | 68.5  | 57.3  | 39.0  |
|                                                                                  | 10%         | WB     | 32.1  | 34.2  | 45.7  | 53.6  | 62.3  | 70.1  | 75.1  | 73.0  | 68.1  | 56.3  | 46.1  | 34.2  |
|                                                                                  |             | MCDB   | 33.9  | 36.1  | 50.7  | 61.2  | 70.9  | 78.5  | 82.1  | 78.9  | 74.9  | 64.0  | 51.4  | 36.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.4  | 16.7  | 16.9  | 20.1  | 20.4  | 19.2  | 19.9  | 19.7  | 22.3  | 20.9  | 18.2  | 15.5  |
|                                                                                  |             | MCDBR  | 18.3  | 19.6  | 26.7  | 29.4  | 27.1  | 23.2  | 21.4  | 21.8  | 25.4  | 27.5  | 25.9  | 19.2  |
|                                                                                  |             | MCWBR  | 15.1  | 15.2  | 16.9  | 15.5  | 13.0  | 10.9  | 11.0  | 11.8  | 12.3  | 14.2  | 16.2  | 15.2  |
|                                                                                  | 5% WB       | MCDBR  | 16.9  | 19.3  | 23.8  | 24.7  | 21.7  | 19.6  | 19.2  | 18.7  | 21.0  | 22.7  | 23.9  | 17.7  |
|                                                                                  |             | MCWBR  | 14.1  | 15.1  | 16.4  | 15.0  | 12.3  | 11.5  | 11.6  | 12.0  | 11.9  | 13.7  | 16.2  | 14.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.264 | 0.277 | 0.307 | 0.358 | 0.378 | 0.397 | 0.408 | 0.405 | 0.362 | 0.321 | 0.282 | 0.264 |
|                                                                                  | taud        |        | 2.450 | 2.426 | 2.453 | 2.344 | 2.353 | 2.357 | 2.310 | 2.292 | 2.390 | 2.496 | 2.528 | 2.470 |
|                                                                                  | Ebn at Noon |        | 279   | 294   | 297   | 287   | 283   | 277   | 272   | 269   | 274   | 273   | 271   | 268   |
|                                                                                  | Edh at Noon |        | 23    | 28    | 31    | 37    | 38    | 38    | 40    | 39    | 33    | 26    | 21    | 20    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 589   | 926   | 1212  | 1496  | 1737  | 1932  | 2078  | 1718  | 1367  | 908   | 605   | 450   |
|                                                                                  | RadStd      |        | 62    | 100   | 116   | 142   | 126   | 140   | 79    | 129   | 95    | 117   | 48    | 44    |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | -1.60 | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (63 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page